



An efficient surrogate-assisted differential evolution algorithm for turbomachinery cascades optimization with more than 100 variables

Zhendong Guo^a, Zijun Zhang^a, Yun Chen^b, Guangjian Ma^b, Liming Song^{a,*}, Jun Li^a, Zhenping Feng^a

^a Xi'an Jiaotong University, Xi'an 710049, China

^b AVIC Shenyang Engine Design Institute, Shenyang 110066, China

ARTICLE INFO

Keywords:

Design optimization
Turbomachinery cascades
High-dimensional expensive black-box problem
More than 100 variables
Surrogate-assisted evolutionary algorithm

ABSTRACT

The CFD-based design optimization of turbomachinery cascades is a typical high-dimensional expensive black-box (HEB) problem. Specifically, to fully consider the interactions between vanes and stages and thus achieving desirable solutions, the cascades of two or more stages have to be optimized simultaneously. The number of variables can reach to more than 100 while the affordable CFD simulations within a design cycle can be only a few hundred or thousand. Hence, how to achieve the optimal design of turbomachinery cascades on budget is very challenging. To solve this problem, a generalized surrogate-assisted differential evolution (DE) algorithm is proposed, which is labeled as GSDE. In GSDE, radial basis function is incorporated into the population regeneration process, which helps to enhance the local exploitation in each iteration and thus achieving a good balance in between global and local search. By validating on benchmark functions ranging from 50 to 100 dimensions, the proposed GSDE algorithm is observed to obtain the true optimal solution with less than 1000 function calls and have far better convergence rate than other state-of-the-art algorithms. Furthermore, GSDE is then tested on the cascades design optimization of a 3.5-stage compressor with 126 variables. Within 1500 CFD evaluations, the total-to-total efficiency is increased by 1.104%. In contrast, classic DE algorithm achieves a similar efficiency gain with nearly 20,000 CFD evaluations. In other words, the computational cost of GSDE amount to only 1/13 of frequently-used evolutionary algorithms. Therefore, the efficiency of the proposed GSDE algorithm for solving high-dimensional turbomachinery optimization problems is well demonstrated.

Introduction

Cascades design optimization is an effective method for further improving the efficiency of turbomachinery [1–3]. Traditional evolutionary algorithms, such as the genetic algorithm (GA) [4], differential evolution (DE) [5], and particle swarm optimization (PSO) [6] are commonly used in conjunction with CFD solvers to guide engineers in the past few decades. However, with the rapid growth of design variables, the requirement of CFD simulations increases exponentially and the problem would soon become very challenging [7,8]. For instance, in order to fully consider the interactions between vanes and blades and that between stages, the cascades of turbomachinery with two or more stages should be optimized simultaneously. Then, more than 100 variables are probably considered in the design space and the number of CFD simulations required for frequently-used evolutionary algorithms would be far more than 10,000. As one CFD simulation of multistage

turbomachinery can take several hours in average, the computational cost of thousands of CFD simulations is tremendous and even prohibitive.

To tackle the above high-dimensional expensive black-box (HEB) optimization problem efficiently, considerable efforts have been devoted in surrogate modeling techniques [9]. Simplified yet accurate and robust models can perform as surrogates to replace part of expensive CFD simulations, thereby enhancing the optimization efficiency. A diversity of surrogates, including polynomial response surface (PRS) [10], Kriging [11–13], radial basis function (RBF) [14,15], support vector regression (SVR) [16,17] and artificial neural network (ANN) [18,19], have been utilized widely in the design optimization of turbomachinery cascades.

Due to the curse of dimensionality, common surrogates can hardly provide accurate approximations with limited sample points, especially when the dimension of the problem is higher than 30 [20]. Thus, surrogate-assisted evolutionary algorithms (SAEAs), which combine

* Corresponding author.

E-mail address: songlm@mail.xjtu.edu.cn (L. Song).

<https://doi.org/10.1016/j.ast.2023.108675>

Received 18 April 2023; Received in revised form 26 September 2023; Accepted 12 October 2023

Available online 17 October 2023

1270-9638/© 2023 Elsevier Masson SAS. All rights reserved.

Nomenclature

CFD	computational fluid dynamics
HEB	high-dimensional expensive black-box
GSDE	generalized surrogate-assisted differential evolution
GA	genetic algorithm
DE	differential evolution algorithm
RBF	radial basis function
SAEAs	surrogate-assisted evolutionary algorithms
GSGA	generalized surrogate-assisted genetic algorithm
y^+	non-dimensional wall distance
η_{is}	isentropic total-to-total efficiency
p_2^0/p_1^0	total pressure ratio
m	mass flow rate
z	axial location of the computational domain
Span	normalized cascade height

surrogate modeling techniques with the typical framework of evolutionary mechanisms, have received enormous attentions in recent years [21,22]. SAEAs not only maintain the advantages of traditional evolutionary algorithms in optimization performance and robustness, but also achieve the convergence acceleration by making full use of the information provided by surrogates. More importantly, since surrogates are only employed as additional tools in SAEAs, the requirement on the accuracy of surrogates is much lower in contrast to surrogate-based algorithms (SBO).

According to the previous studies, there are two popular approaches to take advantage of surrogates in evolutionary algorithms. The first one is the surrogate-based local search. Based on the predicted information provided by surrogates, evolutionary algorithms can significantly strengthen the convergence with little diversity loss in early iterations. Instead of the whole design space, surrogates can also be constructed in local area to achieve higher accuracy [23]. In addition, various strategies such as the trust region method [24] are proposed to further enhance the ability of local search.

The second approach is the prescreening on the surrogate. It often refers to the predicted response of surrogates to choose promising candidates produced by evolutionary operators. The infill sampling criteria based on Kriging model, especially expected improvement (EI) [25] and its variants [26,27], are mostly utilized in the prescreening. Moreover, Gong et al. [28] proposed a strategy based on cheap surrogate model to screen the best candidates generated by different evolutionary operators. Praveen and Duvinneau [29] constructed RBF surrogate for imprecise pre-evaluation to select promising particles in particle swarm optimization.

However, seldom have SAEAs been reported for tackling optimization problems with 100 or more variables so far. The method of how to further integrate surrogate modeling techniques with evolutionary mechanisms is still worth studying. Aiming to accomplish the turbomachinery cascades optimization with more than 100 design variables in less than thousands of CFD simulations, three strategies are integrated in DE with RBF surrogates to improve its optimization efficiency in this paper. To be more specific, a surrogate-guided differential evolution, a parent-based selection and a surrogate-based local search are conducted in parallel in the proposed algorithm, which is called GSDE. The optimum found on the RBF surrogates can lead GSDE search rapidly while a prescreening on the candidates can largely decrease the number of sample evaluations in the optimization process. The superiority of the proposed GSDE algorithm has been demonstrated on four numerical benchmark functions that have 50 to 100 design variables respectively and a design optimization of multistage compressor cascades with 126 variables.

The remainder of the paper is organized as follows. Section 2 gives a

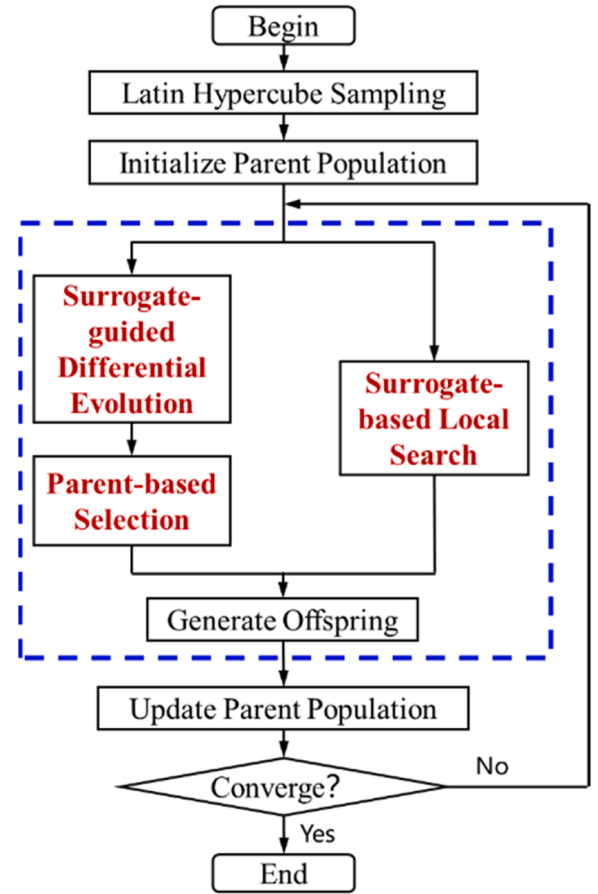


Fig. 1. Flowchart of the GSDE algorithm.

complete explanation about the proposed GSDE algorithm and its performance validation. Section 3 introduces the specific optimization procedure. Section 4 then makes a description on the objective of the cascades design optimization. Section 5 presents the optimization result and aerodynamic analysis. The conclusions are finally drawn in Section 6.

Proposed algorithm

In this paper, we propose a generalized surrogate-assisted differential evolution algorithm in order to deal with the difficulties on CFD-based design optimization of turbomachinery cascades with more than 100 variables. Differential evolution is applied as the basic framework due to its outstanding performance on global optimization. Hence, the proposed algorithm is named as GSDE. RBF is employed as surrogates for local search and population regeneration. The flowchart of the GSDE algorithm is plotted in Fig. 1.

There are three effective strategies, including a surrogate-guided differential evolution, a parent-based selection and a surrogate-based local search, that are employed in GSDE to accelerate the searching process. To be noted, it is a special characteristic of GSDE that the surrogate-guided differential evolution integrates RBF surrogates with differential evolution in a tight form, thus promising the performance of GSDE in high dimensions. Another main contribution of GSDE is the hybrid use of three strategies in parallel, which makes GSDE not only efficient for global optimization but also suitable for the parallel computing environment in real-world problems. As a consequence, the computational cost required for turbomachinery cascades optimization can be greatly reduced by the proposed GSDE algorithm. Without loss of generality, the optimization problems mentioned below are all assumed to be minimization problems.

Radial basis function

Some previous studies [30,31] pointed out that RBF has presented stronger capabilities on the accuracy of prediction than other common surrogates in high-dimensional design space. Moreover, the time consumed in training an RBF remains acceptable when the number of variables increases rapidly. Therefore, the proposed GSDE algorithm uses cubic RBF as surrogates in the optimization process, which is defined as follows.

Suppose that the function values of n distinct training points $x_1, x_2, \dots, x_n \in R^D$ are $f(x_1), f(x_2), \dots, f(x_n)$. The output of the cubic RBF at a new sample point x_{new} has the following expression:

$$\hat{f}(x_{new}) = \sum_{i=1}^n \lambda_i \phi(\|x_{new} - x_i\|) + p(x_{new}) \quad (1)$$

where $\phi(\cdot)$ denotes the cubic basis function that $\phi(r) = r^3$ and λ_i is the weight of the i th cubic basis function. $\|x_{new} - x_i\|$ is the Euclidean distance between x_{new} and x_i .

In addition, it should be noticed that there is an extra linear part $p(x_{new})$ appended in this paper which is a polynomial function:

$$p(x_{new}) = \sum_{i=1}^D b_i x_{new}^i + a \quad (2)$$

The estimation of $n + D + 1$ hyperparameters in the cubic RBF ($\lambda_1, \lambda_2, \dots, \lambda_n \in R, b_1, b_2, \dots, b_D \in R, a \in R$) can be given by the following linear equations:

$$\begin{pmatrix} \Phi & P \\ P^T & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ c \end{pmatrix} = \begin{pmatrix} F \\ 0 \end{pmatrix} \quad (3)$$

where Φ is an $n \times n$ matrix with $\phi(\|x_i - x_j\|)$ to be the (i, j) th element of Φ . P, λ, c, F can be expressed as follows:

$$P = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix}, \lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}, c = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_D \\ a \end{pmatrix}, F = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix} \quad (4)$$

Surrogate-guided differential evolution

Inspired by the idea of previous study [32], a surrogate-guided differential evolution, which combines differential evolution with local RBF surrogates in a tight way, is proposed in GSDE to produce competitive candidates. The predicted optimum on the surrogates can direct GSDE search accurately and quickly while the global search ability is maintained by differential evolution itself.

First, a best/1 differential mutation is conducted for each parent point x_i to generate a mutation point:

$$v_i = x_{best} + F \times (x_{r_1} - x_{r_2}), i = 1, 2, \dots, N \quad (5)$$

where N denotes the population size and it is set as 50 to balance the computational cost of model training and the diversity of population. x_{best} is the best point with minimum fitness value of parent population. r_1^i and r_2^i are two random numbers that satisfy $r_1^i \neq r_2^i \neq i$. F is the mutation operator and its default value is 0.3.

Next, a predicted parent point x_i^* is found in the corresponding neighbor region of each parent point x_i . It is the local optimum on each RBF surrogate constructed with $5 \times D$ training points around each parent point. The radius of the local region around each parent point is set adaptively as:

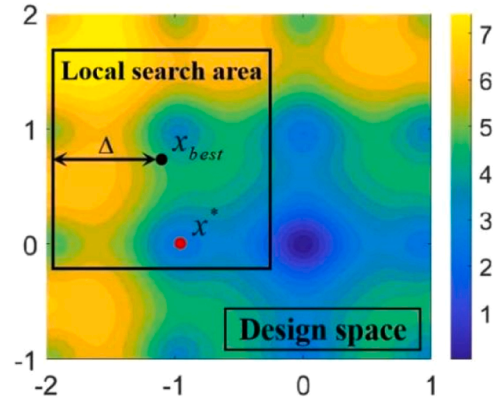


Fig. 2. Surrogate-based local search on 2D design space.

$$r_i = 0.5 \times \frac{D_{i,max}}{\sqrt{D} \sqrt{N-1}}, i = 1, 2, \dots, N \quad (6)$$

where $D_{i,max}$ denotes the maximum distance between x_i and other parent points, and D is the dimension of the problem. Note that, the searching of x_i^* is obtained through optimization with a particle swarm optimization (PSO) algorithm proposed in [6], where the population size of the PSO is set to be 50, and the total number of iterations of PSO is 100. For the local RBF surrogate that used for the searching of x_i^* , it is built as follows. When the number of total training sample points is larger than $5 \times D$, the local RBF surrogate is built by using the $5 \times D$ points that closest to x_i . Otherwise, for the first iteration and the iterations in the very early stage when the total number of evaluated samples is less than $5D$, only one surrogate with all the samples is built for the searching of the x_i^* .

Finally, an exponential crossover operation is adopted between the predicted parent points and mutation points to produce competitive candidates:

$$u_{i,j} = \begin{cases} v_{i,j}, \text{rand}[0, 1] \leq CR \text{ from } j = j_{rand} \\ x_{i,j}^*, \text{otherwise} \end{cases} \quad (7)$$

where $v_{i,j}$ is the j th variable of mutation point v_i . $x_{i,j}^*$ is the j th variable of predicted parent point x_i^* . CR is the crossover operator and its default value is 0.9.

Parent-based selection

In order to further accelerate the convergence rate of GSDE, a parent-based selection method is proposed to prescreen the candidates so that the number of evaluations in each iteration can be decreased a lot. In the procedure of parent-based selection, all candidates u_i are ranked according to the fitness value of their parent population points x_i . Only T candidates with better fitness value will be saved as offspring and then be evaluated exactly, as the candidates with excellent genes are more likely to be considered promising for global optimization. Taking the computational resources into account, $T = 9$ is a proper choice for parallel computing environment of CFD simulations. These offspring will update the parent population by replacing T inferior points with worse fitness values in the parent population.

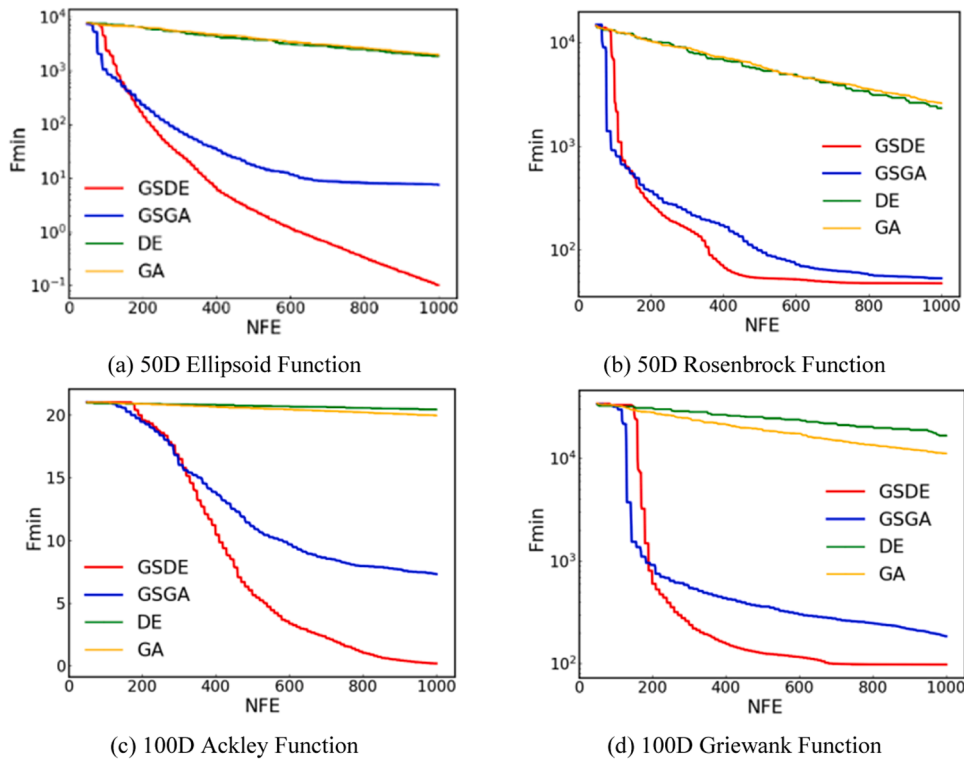
Surrogate-based local search

It has been proved [33,34] that the global optimum provided by surrogates has a strong effect on enhancing the optimization efficiency of evolutionary algorithms. Therefore, similar to Section 2.2, a surrogate-guided local search is carried out near the best point x_{best} with the minimum fitness value of parent population in parallel in GSDE, as depicted in Fig. 2.

Table 1

Benchmark functions and optimization result.

Function	Dimension	Global optimum	Function characteristics	Algorithm	Mean	Std.
Ellipsoid	50	0	Unimodal	GSDE	7.80E-2	5.89E-2
				GSGA	7.22E+0	2.88E+0
				DE	1.70E+3	3.13E+2
				GA	1.86E+3	2.45E+2
Rosenbrock	50	0	Multimodal with narrow valley	GSDE	4.78E+1	6.55E-1
				GSGA	5.18E+1	3.04E+0
				DE	2.20E+3	5.03E+2
				GA	2.51E+3	6.93E+2
Ackley	100	0	Multimodal	GSDE	1.08E-1	7.79E-2
				GSGA	7.12E+0	7.46E-1
				DE	2.04E+1	1.77E-1
				GA	1.99E+1	1.91E-1
Griewank	100	0	Multimodal	GSDE	1.52E-1	5.30E-2
				GSGA	1.01E+0	2.41E-2
				DE	1.44E+3	1.46E+2
				GA	1.21E+3	9.54E+1

**Fig. 3.** Mean convergence curves on different benchmark functions.

First, a local RBF surrogate is constructed around x_{best} . Training points that are of five times of problem dimension are selected to improve its accuracy. In the early stages when the number of evaluated samples is less than five times of the problem dimension, all evaluated sample points will be used in the training process. After that, a PSO algorithm is applied to find the local optimum x^* in the small area of local RBF surrogate:

$$x^* = \operatorname{argmin}(\hat{f}_{RBF}(x)), x \in [x_{best} - \Delta, x_{best} + \Delta] \cap [lb, ub] \quad (8)$$

where Δ denotes the radius of local search area and it is set as half the distance between x_{best} and the point with maximum fitness value among training points. lb and ub denote the lower bound and upper bound of the design space respectively, and x^* is the offspring generated by surrogate-based local search and it will replace the current best solution x_{best} in the parent population if the fitness value of it is smaller than that of x_{best} .

Performance validation

In order to validate the performance of the proposed GSDE algorithm, four classic numerical functions with dimensions varying from 50 to 100 are selected as the benchmark problems. In addition to the classic evolutionary algorithms such as DE [5] and GA [4], the state-of-the-art surrogate-assisted evolutionary algorithm GSGA [32] is selected as a baseline to compare with GSDE. The design space of the benchmark functions are all normalized to [0, 1] and the maximum number of function calls are all set to be 1000. For fair comparison, 10 runs are carried out for each algorithm on each benchmark functions. The information about the benchmark functions and the statistics of the optimization result are listed in Table 1.

It can be found that the proposed GSDE algorithm can always obtain an excellent solution near the global optimum when testing the four benchmark functions. Compared with other algorithms, GSDE can always achieve the best optimal solution, and the performance

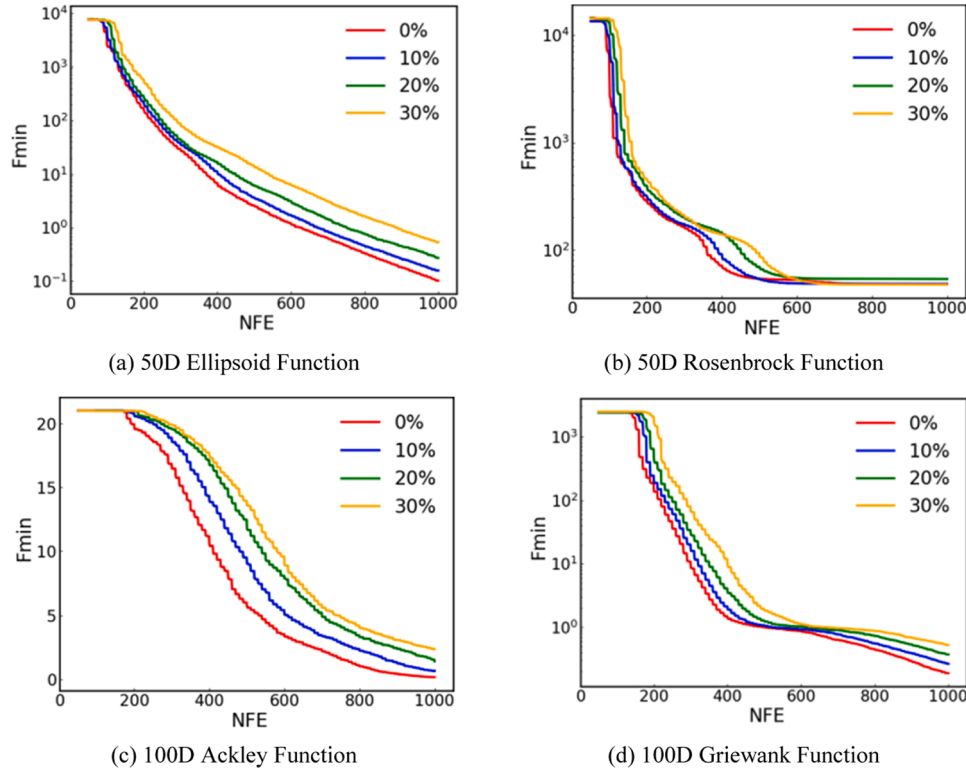


Fig. 4. Robustness validation of GSDE with different probability of CFD evaluation failures.

fluctuations that represented by the standard deviation value are always the smallest. To show the advantage of GSDE more clearly, the mean convergence curves of 10 runs are plotted in Fig. 3, where the horizontal axis shows the number of function evaluations consumed in the optimization process, denoted by NFE; and the vertical coordinate presents the related current best solution for minimization problems, represented by Fmin. Obviously, when comparing to the classic evolutionary algorithms DE and GA, the surrogate-assisted evolutionary algorithms such as GSGA and GSDE are shown to have far better convergence rates, as the slope of the mean convergence curves of GSGA and GSDE are much sharper than those of DE and GA in Fig. 3. The final solutions and the standard deviation of the optimization results of GSGA and GSDE also look better than those of DE and GA, as shown in both Fig. 3 and Table 1. When comparing GSDE with GSGA, the final solutions of GSDE are shown to be better than those of GSGA by one or two order of magnitude on benchmark functions of Ellipsoid, Ackley and Griewank as shown in Table 1. In the meantime, when the GSDE achieving the similar optimized solutions as the final solutions of GSGA, the number of function evaluations of GSDE is just about half of the GSGA. With the above, the effectiveness of proposed GSDE algorithm has been well demonstrated.

Moreover, it is often the case that part of the CFD evaluations may fail in the middle of the optimization process, which can directly lead to the fail of the turbomachinery cascade optimization. Hence, dealing with CFD computation fails is of great importance. Since GSDE generates multiple offspring in one iteration, the samples with failed CFD evaluations are simply abandoned. To validate the robustness of GSDE, we set a probability of computational fails to when testing functions. For instance, 30% means 30% sample evaluations by CFD would be failed in each iteration. As shown in Fig. 4, with the increasing probability of CFD evaluation failures, the performance of GSDE algorithm almost remained the same as the case without CFD evaluation failures. Also note that, those failed samples are also included in the number of evaluations. That is to say, GSDE is robust enough against CFD evaluation failures.

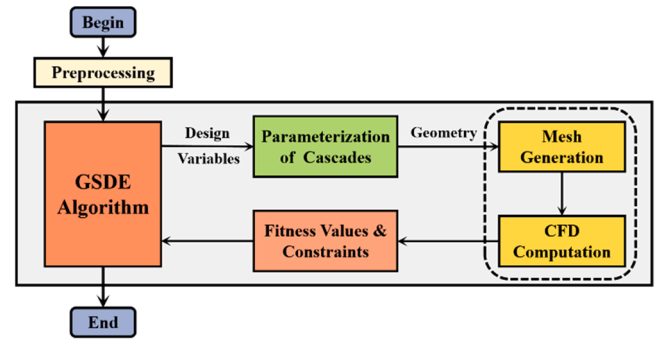


Fig. 5. Optimization procedure.

Optimization procedure of turbomachinery cascades

The design optimization of turbomachinery cascades is implemented in C++ language with an inhouse code. Fig. 5 shows the flowchart of the optimization procedure. There are four important modules, including the GSDE algorithm, the parameterization method, the CFD-based performance evaluation and the constraint handling method. The detailed steps of the optimization procedure for the cascade design is described as follows:

Step1: To begin with, the settings of the cascades design optimization and the templates of mesh generation and CFD aerodynamic analysis are prepared in preprocessing.

Step2: Promising sample points are produced by the proposed GSDE algorithm with their design variables.

Step3: Non-Uniform Rational B-Spline (NURBS) curves is employed as the parameterization method of 3D blade to construct the geometry of turbomachinery cascades.

Step4: CFD-based performance evaluations are conducted on the sample points according to the templates.

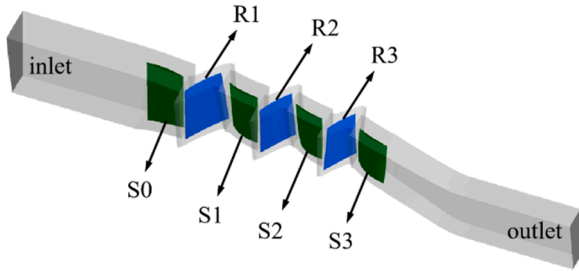


Fig. 6. Computational domain.

Step5: A penalty function is programmed to check whether the constraints are satisfied. Then, the fitness values of the sample points are collected from the computational result and input into the GSDE algorithm. If the terminal condition is met, the current optimal design coupled with its aerodynamic performance is output as the optimization result, otherwise return to Step2.

Design optimization of compressor cascades

Geometry definition

In this paper, the cascades of a compressor with 3.5 stages are adopted as the research object to demonstrate the effectiveness of the proposed GSDE algorithm on turbomachinery cascades optimization with more than 100 variables. Fig. 6 shows the computational domain of the compressor, including a row of inlet guide vanes (S0) and three low-pressure compressor stages (R1&S1, R2&S2, R3&S3). In the CFD simulations, it is supposed that the stators S0, S1, S2 and S3 have no tip clearance while the clearance at the tip of the rotors R1, R2 and R3 are all fixed at 0.2 mm.

Blade parameterization

The parameterization method of the 3.5-stage compressor cascades is programmed based on NURBS curves [35] to determine its design space. To begin with, three 2D sections, including the root, middle and tip section of the compressor cascades, are considered to constitute the profile of each blade. Since the suction surface of the blade usually has much stronger influence on the aerodynamic losses than the pressure surface, only the suction surface of the sections is going to be adjusted.

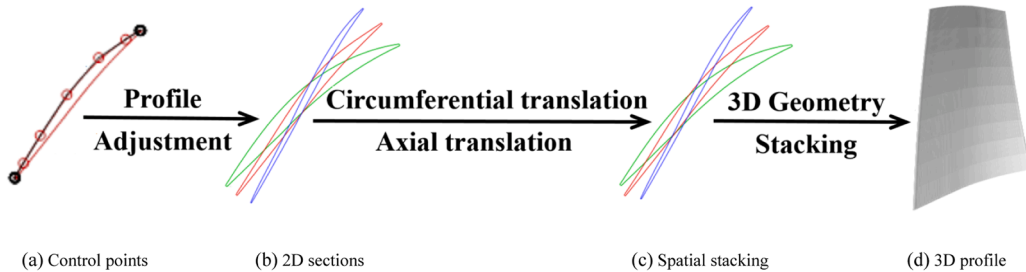


Fig. 7. Parameterization method.

Specifically, a total number of seven control points are determined on the suction surface of each section, in which two of them are fixed at the leading and trailing edge while the other five are active along the normal direction of fitted B-Spline curve (see Fig. 7(a)). Each active control point can be described by one parameter and therefore fifteen design variables are determined for each blade.

After the parameterization of the sections, three extra design variables are employed to adjust the lean and sweep of each blade. To be more specific, two parameters are set to represent the circumferential translation of both the middle and tip sections, and one parameter is set to represent the axial translation of the tip section (see Fig. 7(b) and (c)). Finally, a 3D profile of each blade is obtained by stacking three 2D sections with skinning surface techniques (see Fig. 7(d)).

By using the blade parameterization method shown in Fig. 7, Table 2 shows the parameterization variables for a blade of the 3.5-stage compressor cascades. Specifically, 18 design variables are used to parameterize each blade, and 126 variables in total are used for the parameterization of the 3.5-stage compressor cascades with 7 blade rows.

Numerical method

To evaluate the aerodynamic performance of the compressor, the commercial CFD software NUMECA FINE/Turbo is adopted as the solver of three-dimensional steady Reynolds average Navier-Stokes equations. To take the turbulence effects into account, the Spalart-Allmaras model is applied as the turbulence model. High-quality hexahedral-structured grids are generated around each blade with O4H mesh topology and a total number of 3.22 million nodes are determined by the investigation of grid independence. The grid height of the first layer near the wall is

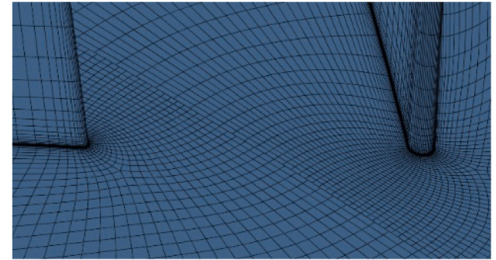


Fig. 8. Computational grids.

Table 2

Parameterization variables for a blade of the 3.5-stage compressor cascades.

S0 (R1/S1/R2/S2/R3/S3)							
Section	Profile control point (from leading to trailing edge)					Circumferential translation	Axial translation
root	x_1	x_2	x_3	x_4	x_5	/	/
middle	x_6	x_7	x_8	x_9	x_{10}	x_{16}	/
tip	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{17}	x_{18}
Range of variation (axial chord length)	$\pm 0.6\%$	$\pm 1.2\%$	$\pm 1.5\%$	$\pm 1.2\%$	$\pm 0.6\%$	$\pm 15\%$	$\pm 5\%$

Table 3
Optimization result of the compressor.

Parameter	Reference	GSDE (1000)	GSDE (1500)	DE
η_{is} (%)	85.825	86.867	86.929	86.940
Increment (%)	–	1.042	1.104	1.115
p'_2/p'_1	1.7534	1.7522	1.7521	1.7533
Relative Variation (%)	–	–0.068	–0.074	-5.7×10^{-3}
m (kg/s)	30.18	30.27	30.28	30.38
Relative Variation (%)	–	+0.30	+0.33	+0.66
Number of CFD simulations	–	1000	1500	20,000

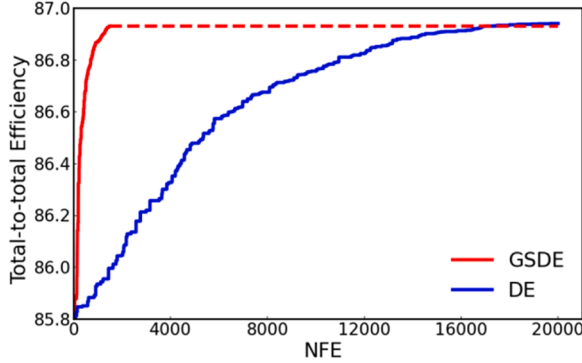


Fig. 9. Convergence curves of total-to-total efficiency.

set as 0.001 mm, thus ensuring the y^+ value is less than 1.0. Details of the computational grids are illustrated in Fig. 8.

On the purpose of verifying the efficiency of the proposed GSDE algorithm, only the design condition of the compressor is considered for the optimization. In the CFD simulations, the inlet boundary condition is selected as an absolute total temperature of 288.15 K and an absolute total pressure of 101,325 Pa, and the average static pressure given at the outlet is 164,000 Pa. The rotational speed of the rotors R1, R2 and R3 is 8297.7 RPM. The surface of the cascades and the hub and shroud of the passage are all set as adiabatic no-slip conditions.

Result analysis

Optimization result

The isentropic total-to-total efficiency of the 3.5-stage compressor is selected to be the optimization objective. Two constraints are set to limit the relative variation of both total pressure ratio and mass flow rate:

$$\max F_{obj}(x) = \eta_{is}(x)/p(x)$$

$$s.t. 0.98(p'_2/p'_1)_{ref} \leq p'_2/p'_1(x) \leq 1.05(p'_2/p'_1)_{ref} \quad (9)$$

$$0.98m_{ref} \leq m(x) \leq 1.02m_{ref}$$

where x denotes the value of design variables. η_{is} is the isentropic total-to-total efficiency and $p(x)$ is the penalty function [2,36] adopted for handling the constraints. p'_1 and p'_2 denote the absolute total pressure at the inlet and outlet respectively. p'_2/p'_1 is the total pressure ratio accordingly. m denotes the mass flow rate. The subscript of ref refers to the value of reference design.

To further show the effectiveness of proposed GSDE algorithm, the cascade design optimizations are carried out by using GSDE and DE, respectively, with Eq. (9). Considering the fact that the coupling of the codes of optimization algorithm with those of blade parameterization method and CFD-based solver technique is not a trivial work, particularly when parallel computation of several CFD samples should be involved to accelerate the optimization progress, therefore GSGA and GA are not considered for the cascade design optimization.

Table 3 shows the optimization results of GSDE and DE, and Fig. 9 shows the convergence curves of GSDE and DE. Specifically in Table 3, GSDE(1000) denotes the optimization results when 1000 CFD evaluations are consumed, and GSDE(1500) lists the optimization results when 1500 CFD evaluations are consumed. Obviously, while achieving similar optimization result as DE, the number of CFD evaluations required by GSDE is significantly reduced. Similar to the validations on benchmark functions (see Fig. 3), the slope of the convergence curve of GSDE looks much sharper than that of DE, indicating a far better convergence rate of GSDE. More specifically, the compressor cascade design is carried out on a micro server with two AMD EPYC 7502 32-Core Processor (2.50 GHz) and 256 GB RAM, and one CFD evaluation of a compressor cascade design would take 1~1.5 h, and 10 CFD samples will be calculated in parallel in each iteration of GSDE. Thereby, the time cost of GSDE for the cascade design optimization with 126 variables takes about one week. In contrast, DE needs 20,000 CFD evaluations to achieve similar results, and the time cost of DE for finishing the same task would take more than two month. In other words, the design efficiency of turbomachinery cascades design can be greatly improved.

Aerodynamic analysis

To figure out the reason why the isentropic total-to-total efficiency can be improved, an aerodynamic analysis is carried out to compare the optimal design obtained by GSDE with the reference design. To begin with, Fig. 10 displays the geometry comparison of the compressor cascades.

Significant differences can be observed on each blade. For the inlet guide vane S0, the blade leans towards the suction side and sweeps

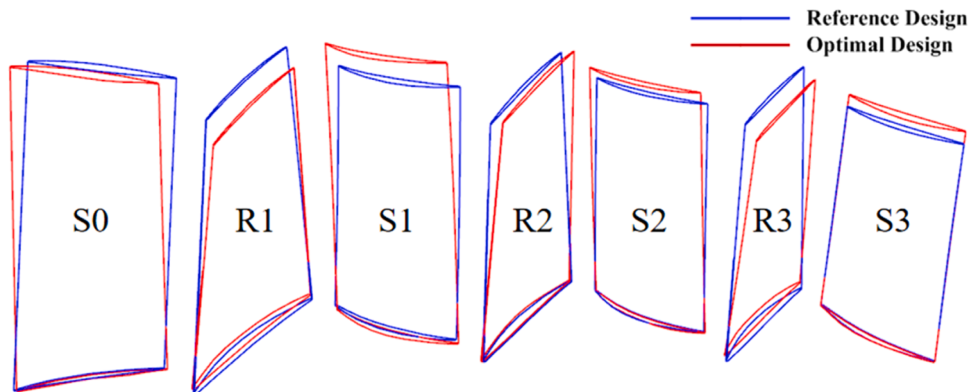


Fig. 10. Comparison of cascades geometry.

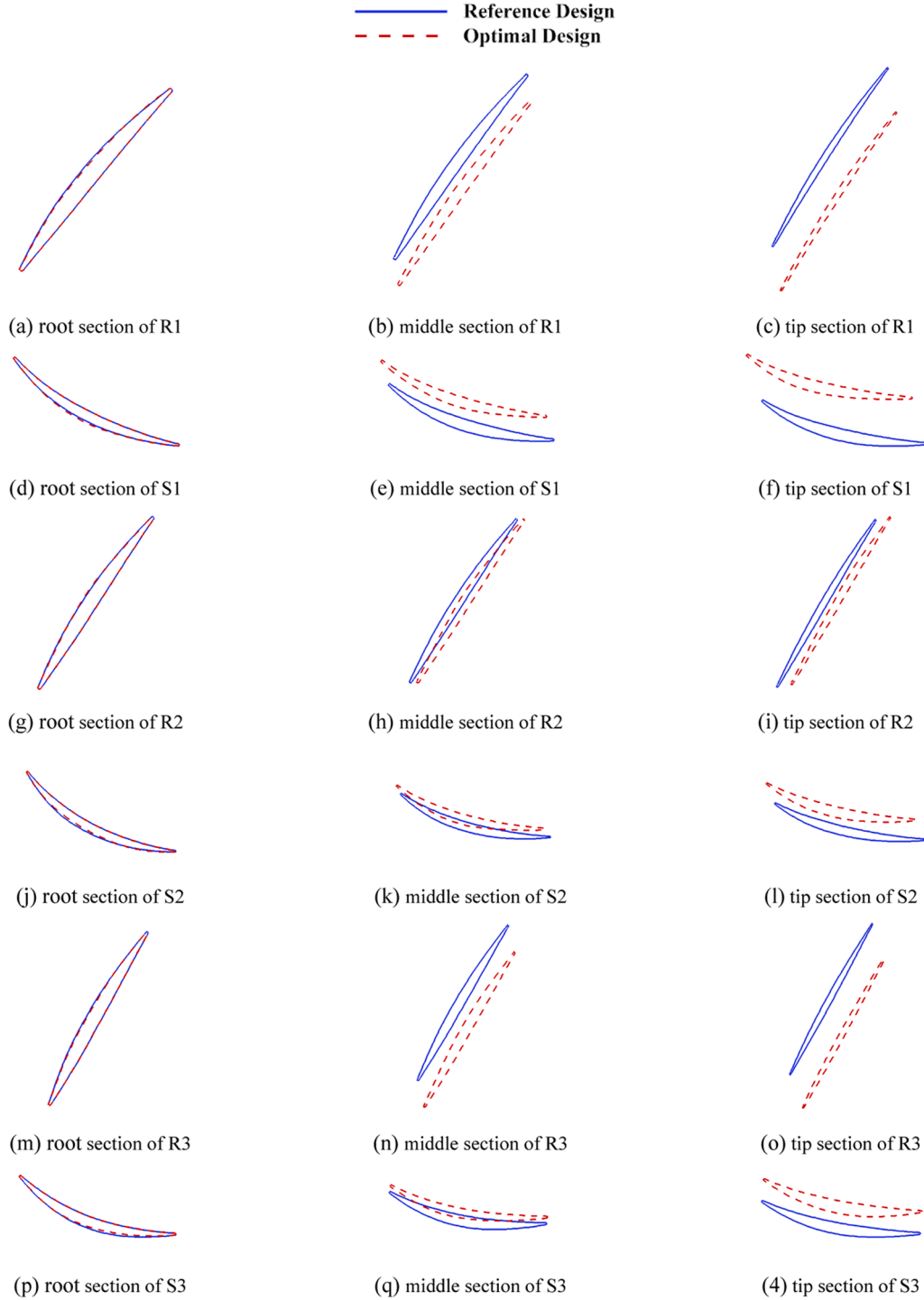


Fig. 11. Comparison of section profiles.

backward. For other three low-pressure compressor stages, however, the blades all lean towards the pressure side. Moreover, the rotors R1, R2 and R3 all sweep forward and the stators S1, S2 and S3 all sweep backward. To make further analysis, the section profiles of different blades are then plotted in Fig. 11. Except S0, other six blades are all included. Although it is not very obvious, we can still observe that the profiles on the suction surface of the blades are all adjusted and most of them have a tendency to become thinner (see Fig. 11(1a), (4a), (5a), (6a)). Moreover, the profiles of the middle section and the tip section are strongly affected by circumferential and axial translation. The direction of circumferential translation is towards the pressure side of the blades

while the direction of axial translation is not regular.

After that, Fig. 12 presents the distribution of isentropic Mach number on the section profiles of all seven blades at different spans. We can see that it is almost identical between the reference design and optimal design on the stators S0, S1, S2, and S3. To be noted, the fluid near the stators flows at subsonic speed, which only accounts for a small part of the aerodynamic losses. As for rotors R1, R2 and R3, the isentropic Mach number has an obvious increase with the increase of the blade height. Supersonic regions appear at 50% and 90% spans near the suction surface of R1 and 90% span near the suction surface of R2. The isentropic Mach number of these regions fluctuates greatly due to the

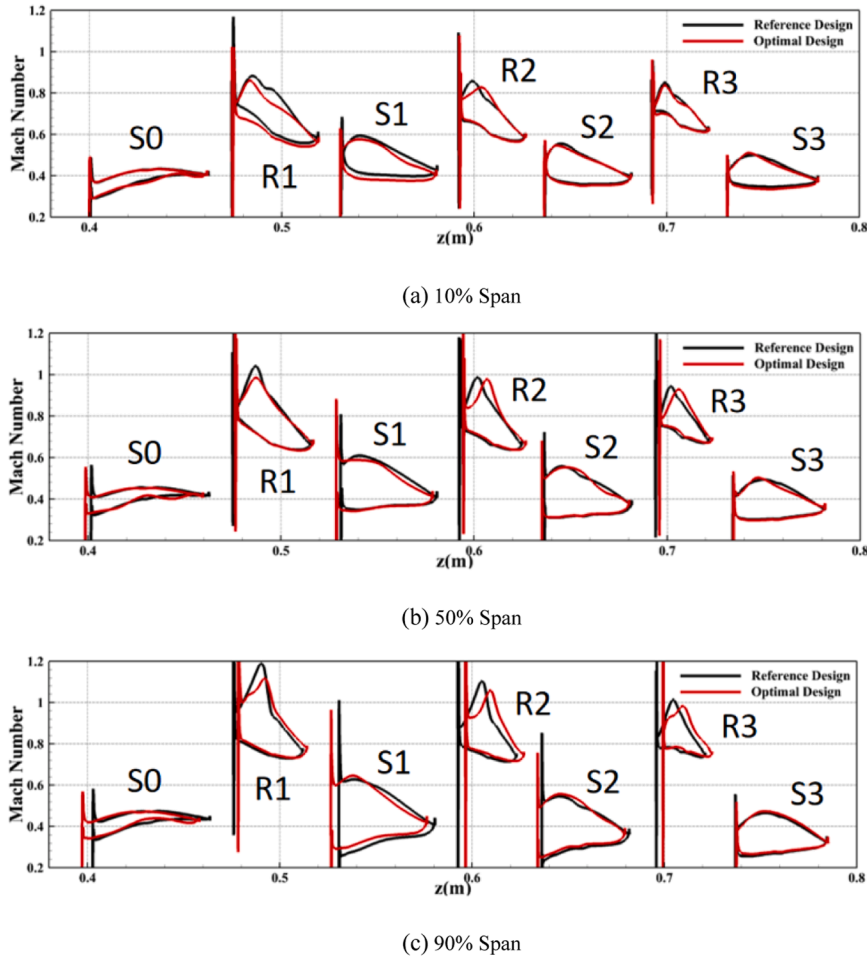


Fig. 12. Distribution of isentropic Mach number on cascades profiles.

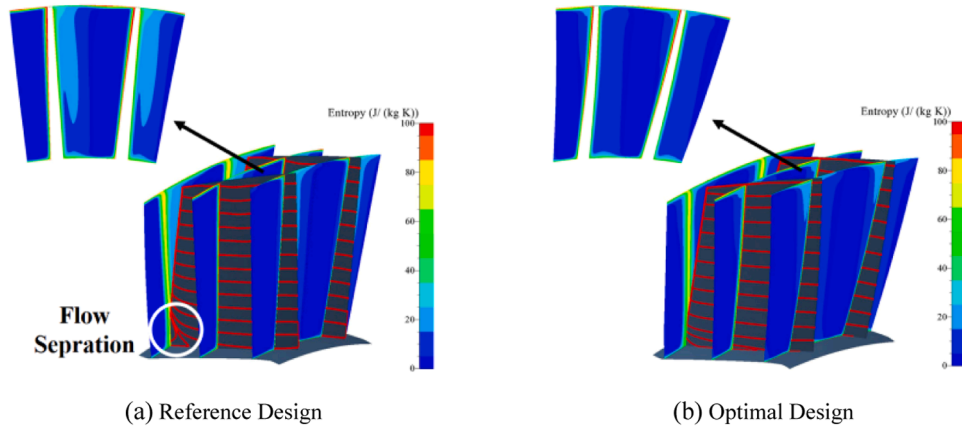


Fig. 13. Streamlines and entropy distribution of R1.

shock wave. Since the optimal design makes only a small adjustment on the profiles of the root section without circumferential and axial translations, a slight decrease can be found on the isentropic Mach number of R1 and R2 at 10% span. At 50% span, the isentropic Mach number on the suction surface of R1 is decreased and smoothened while the peak of isentropic Mach number on the suction surface of R2 and R3 is moved forward. The comparison at 90% span is similar to 50% span with an additional decrease on the isentropic Mach number of R2 and R3. On a whole, the supersonic flow in the passage of the rotors is weakened by

the optimal design, thereby largely reducing the aerodynamic losses.

Since the fluid flow near R1 and R2 seems to be obviously affected, the characteristics in the flow field of R1 and R2 are compared. Fig. 13 shows the limiting streamlines on the suction surface of R1 and entropy distribution in the passage. For reference design, the flow separation at the corner of trailing edge can be clearly observed as marked in the white circle. The regions with high entropy appear near the pressure surface of the blade. After optimization, the limiting streamlines become flatter and the entropy is decreased, which means the fluid flows more

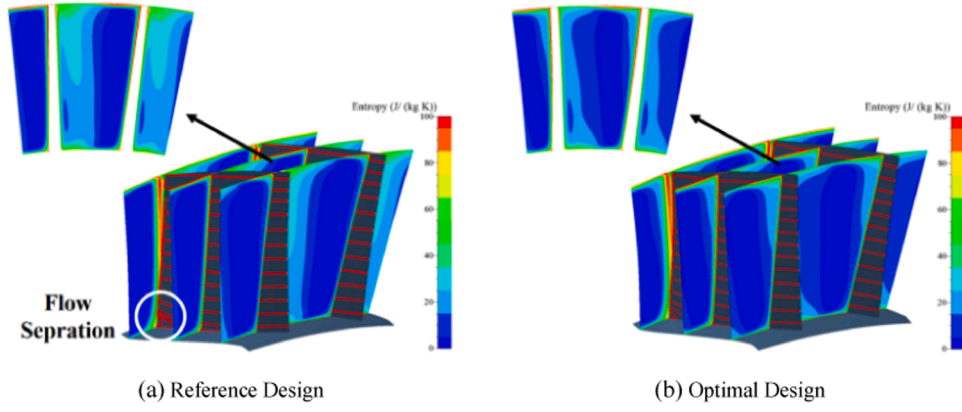


Fig. 14. Streamlines and entropy distribution of R2.

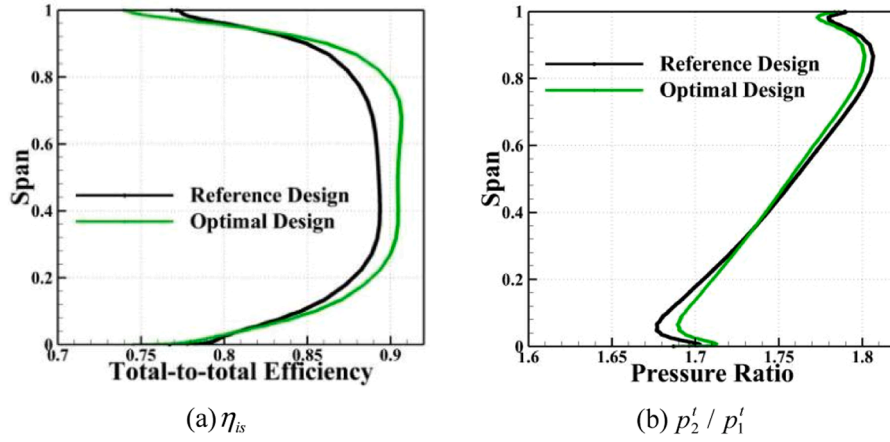


Fig. 15. Spanwise distribution of overall performance.

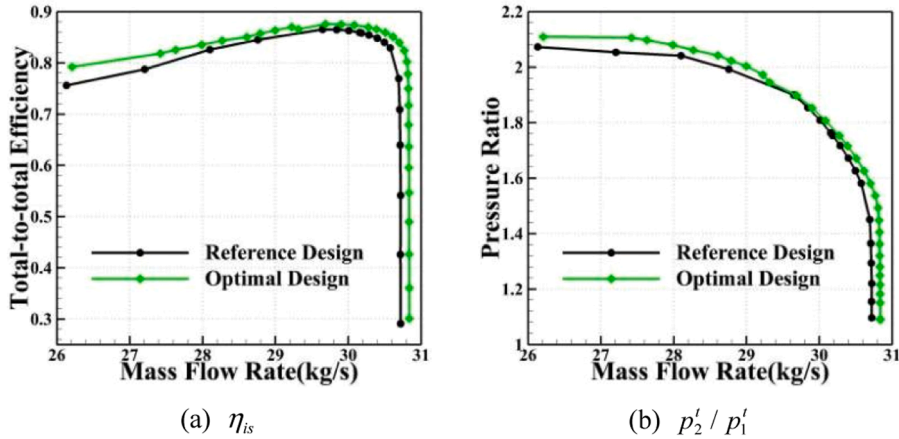


Fig. 16. Performance under varying conditions.

stably in the passage.

Similarly, the limiting streamlines and entropy distribution of R2 are illustrated in Fig. 14. The improvement on the flow separation of R2 is the same as that of R1. For aerodynamic losses, the optimal design not only reduces the entropy in the passage but also makes it uniform between the pressure surface and the suction surface.

Finally, Fig. 15 compares the spanwise distribution of isentropic total-to-total efficiency and total pressure ratio between the optimal design and reference design. We can see that the isentropic total-to-total

efficiency is obviously increased for the optimal design over the span range from 5% to 95%. Meanwhile, the total pressure ratio of the optimal design is increased over 0% to 35% area and decreased over 35% to 100% area of the span. As a result, the total pressure ratio is distributed evenly along the span after the optimization.

Performance under varying conditions

Since the compressor often works in a wide range apart from the

design condition, the feature curves at off-design conditions are crucial for its stability. The isentropic total-to-total efficiency and total pressure ratio of both the optimal design and reference design under varying conditions are compared in Fig. 16 for performance check. It is indicated that the overall performance of the optimal design is improved in different degrees throughout the entire working range. To be noticed, the surge margin is also expanded for the optimal design. In general, the optimal design is demonstrated to be significantly better than the reference design on the aerodynamic performance at both design and off-design conditions.

Conclusion

A generalized surrogate-assisted differential evolution algorithm is proposed and programmed in this paper for turbomachinery cascades optimization with 100 or more variables, which is named as GSDE. The performance of GSDE has been tested on 50D and 100D numerical benchmark functions. After that, an aerodynamic design optimization is carried out on the cascades of a 3.5-stage compressor which have 126 variables. Some conclusions are drawn as follows:

- (1) The proposed GSDE algorithm is composed of RBF surrogates, a surrogate-guided differential evolution, a parent-based selection and a surrogate-based local search. Taking parallel computing environment into account, the sample evaluations in each iteration are conducted in parallel in GSDE. By testing on high-dimensional unimodal and multimodal benchmark functions, the proposed GSDE algorithm shows the capability of obtaining global optimum with less than 1000 evaluations, which is shown to have better convergence rate and can obtain better final solutions than the other compared algorithms.
- (2) By combining GSDE with 3D blade parameterization method, RANS solver technique and penalty function based constraint handling method, an optimization is carried out for maximizing the total-to-total efficiency of the 3.5 stage compressor. After optimization, the total-to-total efficiency of the compressor is increased by 1.104%, which is very close to the optimization result of classic DE algorithm. However, only 1500 CFD evaluations are required for GSDE, which is only 1/13 of classic DE. When comparing to the reference design, the optimal design obtained by GSDE is indicated to be helpful to weaken the supersonic flow in the passage and the corner separation on the suction side of the rotors according to detailed aerodynamic analysis.

In view of the above, the proposed GSDE algorithm is demonstrated to be efficient for turbomachinery optimization with more than 100 variables.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The author would like to acknowledge the support from the National Major Science and Technology Project [2019-II-0008-0028], the Industry-University-Research Cooperation Project of Aero Engine Corporation of China [HFZL2021CX004], the High-level Innovative and

Entrepreneurial Talents Introduction Project of Qinchuangyuan (QCYRCXM-2022-210), and the National Science Foundation of China (No.52306048).

References

- [1] J. Luo, J. Xiong, F. Liu, I. McBean, Three-dimensional aerodynamic design optimization of a turbine blade by using an adjoint method, *J. Turbomach.* 133 (1) (2010), 011026.
- [2] C. Luo, L. Song, J. Li, Z. Feng, A study on multidisciplinary optimization of an axial compressor blade based on evolutionary algorithms, *J. Turbomach.* 134 (5) (2012), 054501.
- [3] C. Li, J. Wang, Z. Guo, L. Song, J. Li, Aero-mechanical multidisciplinary optimization of a high speed centrifugal impeller, *Aerosp. Sci. Technol.* 95 (2019), 105452.
- [4] M. Mitchell, *An Introduction to Genetic Algorithms*, MIT press, 1998.
- [5] R. Storn, K. Price, Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces, *J. Glob. Optim.* 11 (4) (1997) 341–359.
- [6] R. Eberhart, J. Kennedy, A new optimizer using particle swarm theory, in: *Proceedings of the MHS'95, Sixth International Symposium on Micro Machine and Human Science*, 1995, pp. 39–43.
- [7] R. Schnell, J. Yin, C. Voss, E. Nicke, Assessment and optimization of the aerodynamic and acoustic characteristics of a counter rotating open rotor, in: *Proceedings of the ASME Turbo Expo: Power for Land, Sea, and Air*, 2010, pp. 11–25.
- [8] M. Aulich, U. Siller, High-dimensional constrained multiobjective optimization of a fan stage, in: *Proceedings of the ASME Turbo Expo: Power for Land, Sea, and Air*, 2011, pp. 1185–1196.
- [9] G.G. Wang, S. Shan, Review of metamodeling techniques in support of engineering design optimization, *J. Mech. Des.* 129 (4) (2007) 415–426.
- [10] R.F. Gunst, Response surface methodology: process and product optimization using designed experiments, *Technometrics* 38 (3) (1996) 284–286.
- [11] L. Song, Z. Guo, J. Li, Z. Feng, Research on metamodel-based global design optimization and data mining methods, *J. Eng. Gas Turbine Power* 138 (9) (2016), 092604.
- [12] D. Zhan, Y. Meng, H. Xing, A fast multi-point expected improvement for parallel expensive optimization, *IEEE Trans. Evol. Comput.* 27 (1) (2022) 170–184.
- [13] D. Ba, Q. Zhang, J. Du, Z. Li, H. Zhang, C. Nie, Design optimization of axial slot casing treatment in a highly-loaded mixed-flow compressor, *Aerosp. Sci. Technol.* 107 (2020), 106262.
- [14] Y.S. Ong, P.B. Nair, A.J. Keane, Evolutionary optimization of computationally expensive problems via surrogate modeling, *AIAA J.* 41 (4) (2003) 687–696.
- [15] R.G. Regis, Evolutionary programming for high-dimensional constrained expensive black-box optimization using radial basis functions, *IEEE Trans. Evol. Comput.* 18 (3) (2014) 326–347.
- [16] H. Hu, J. Yu, Y. Song, F. Chen, The application of support vector regression and mesh deformation technique in the optimization of transonic compressor design, *Aerosp. Sci. Technol.* 112 (2021), 106589.
- [17] H. Yuan, M. Wang, J. Zhang, Y. Zhang, X. Lu, Integrated optimization of a high-lift low-pressure turbine cascade based on dynamic support vector regression, *Aerosp. Sci. Technol.* 131 (2022), 107986.
- [18] D.I. Lopez, T. Ghisu, S. Shahpar, Global optimization of a transonic fan blade through AI-enabled active subspaces, *J. Turbomach.* 144 (1) (2022), 011013.
- [19] S.B. Ma, K.Y. Kim, Stability enhancement of a centrifugal compressor using inclined discrete cavities, *Aerosp. Sci. Technol.* 107 (2020), 106252.
- [20] D. Zhan, H. Xing, A fast kriging-assisted evolutionary algorithm based on incremental learning, *IEEE Trans. Evol. Comput.* 25 (5) (2021) 941–955.
- [21] Y. Jin, Surrogate-assisted evolutionary computation: recent advances and future challenges, *Swarm Evol. Comput.* 1 (2) (2011) 61–70.
- [22] R.T. Haftka, D. Villanueva, A. Chaudhuri, Parallel surrogate-assisted global optimization with expensive functions—a survey, *Struct. Multidiscip. Optim.* 54 (1) (2016) 3–13.
- [23] Z. Zhou, Y.S. Ong, P.B. Nair, A.J. Keane, K.Y. Lum, Combining global and local surrogate models to accelerate evolutionary optimization, *IEEE Trans. Syst. Man Cybern. C Appl. Rev.* 37 (1) (2006) 66–76.
- [24] T. Long, X. Li, R. Shi, J. Liu, X. Guo, L. Liu, Gradient-free trust-region-based adaptive response surface method for expensive aircraft optimization, *AIAA J.* 56 (2) (2018) 862–873.
- [25] D.R. Jones, M. Schonlau, W.J. Welch, Efficient global optimization of expensive black-box functions, *J. Glob. Optim.* 13 (4) (1998) 455–492.
- [26] Z. Guo, Q. Wang, L. Song, J. Li, Parallel multi-fidelity expected improvement method for efficient global optimization, *Struct. Multidiscip. Optim.* 64 (3) (2021) 1457–1468.
- [27] T. Long, Z. Wei, R. Shi, Y. Wu, Parallel adaptive kriging method with constraint aggregation for expensive black-box optimization problems, *AIAA J.* 59 (9) (2021) 3465–3479.
- [28] W. Gong, A. Zhou, Z. Cai, A multioperator search strategy based on cheap surrogate models for evolutionary optimization, *IEEE Trans. Evol. Comput.* 19 (5) (2015) 746–758.
- [29] C. Praveen, R. Duvinneau, Low cost PSO using metamodels and inexact pre-evaluation: application to aerodynamic shape design, *Comput. Methods Appl. Mech. Eng.* 198 (9–12) (2009) 1087–1096.
- [30] A.A. Mullur, A. Messac, Metamodeling using extended radial basis functions: a comparative approach, *Eng. Comput.* 21 (3) (2006) 203–217.

- [31] A. Díaz-Manríquez, G. Toscano, C.A Coello Coello, Comparison of metamodeling techniques in evolutionary algorithms, *Soft Comput.* 21 (19) (2017) 5647–5663.
- [32] X. Cai, L. Gao, X. Li, Efficient generalized surrogate-assisted evolutionary algorithm for high-dimensional expensive problems, *IEEE Trans. Evol. Comput.* 24 (2) (2019) 365–379.
- [33] R.G. Regis, Particle swarm with radial basis function surrogates for expensive black-box optimization, *J. Comput. Sci.* 5 (1) (2014) 12–23.
- [34] H. Yu, Y. Tan, J. Zeng, C. Sun, Y. Jin, Surrogate-assisted hierarchical particle swarm optimization, *Inf. Sci.* 454-455 (2018) 59–72. Ny.
- [35] Z. Guo, H. Bu, L. Song, J. Li, Z. Feng, Experimental test of a 3D parameterized vane cascade with non-axisymmetric endwall, *Aerosp. Sci. Technol.* 85 (2019) 429–442.
- [36] M. Gen, Interval programming using genetic algorithms, in: *Proceedings of the 12th Fuzzy System Symposium*, 1996, 1996.